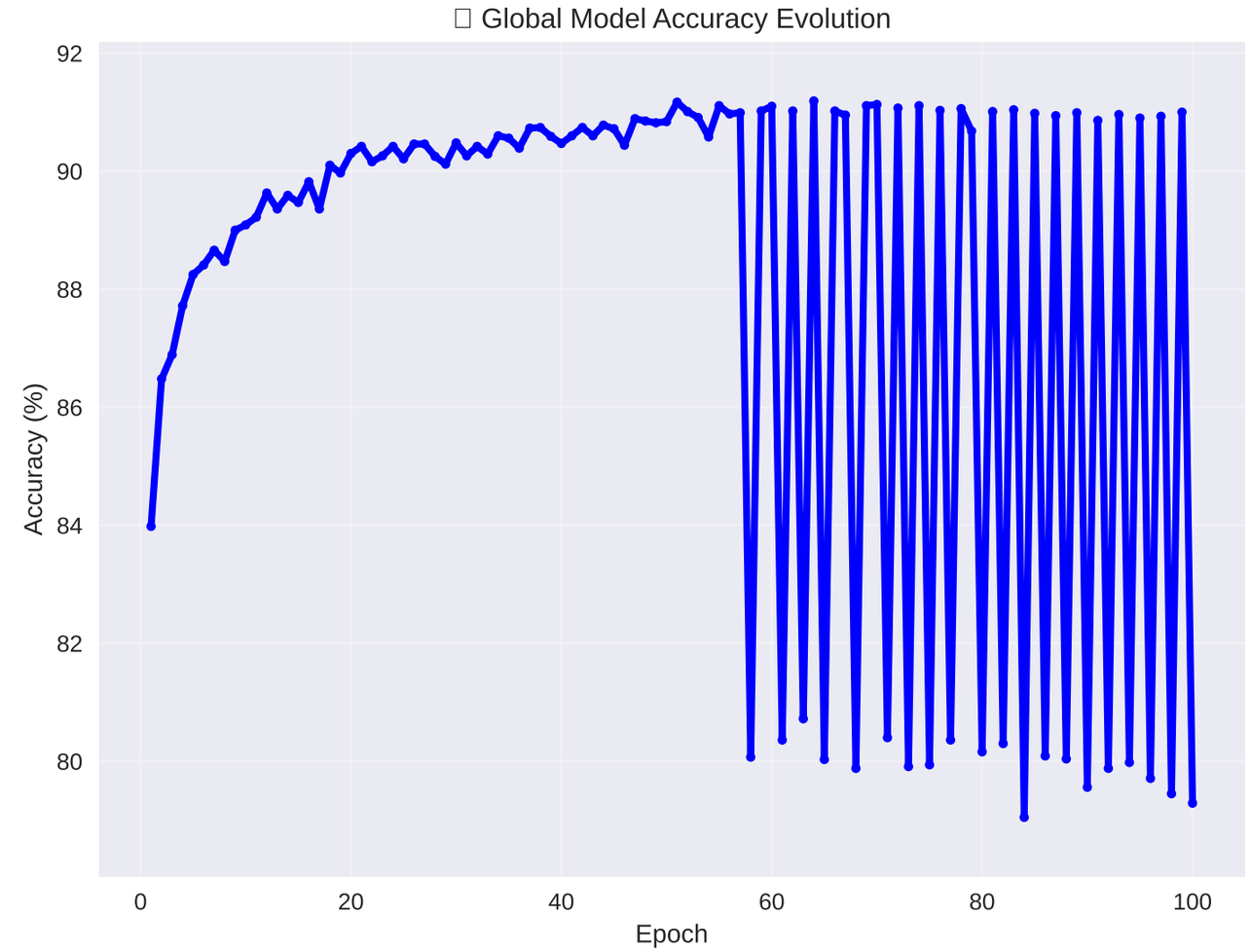
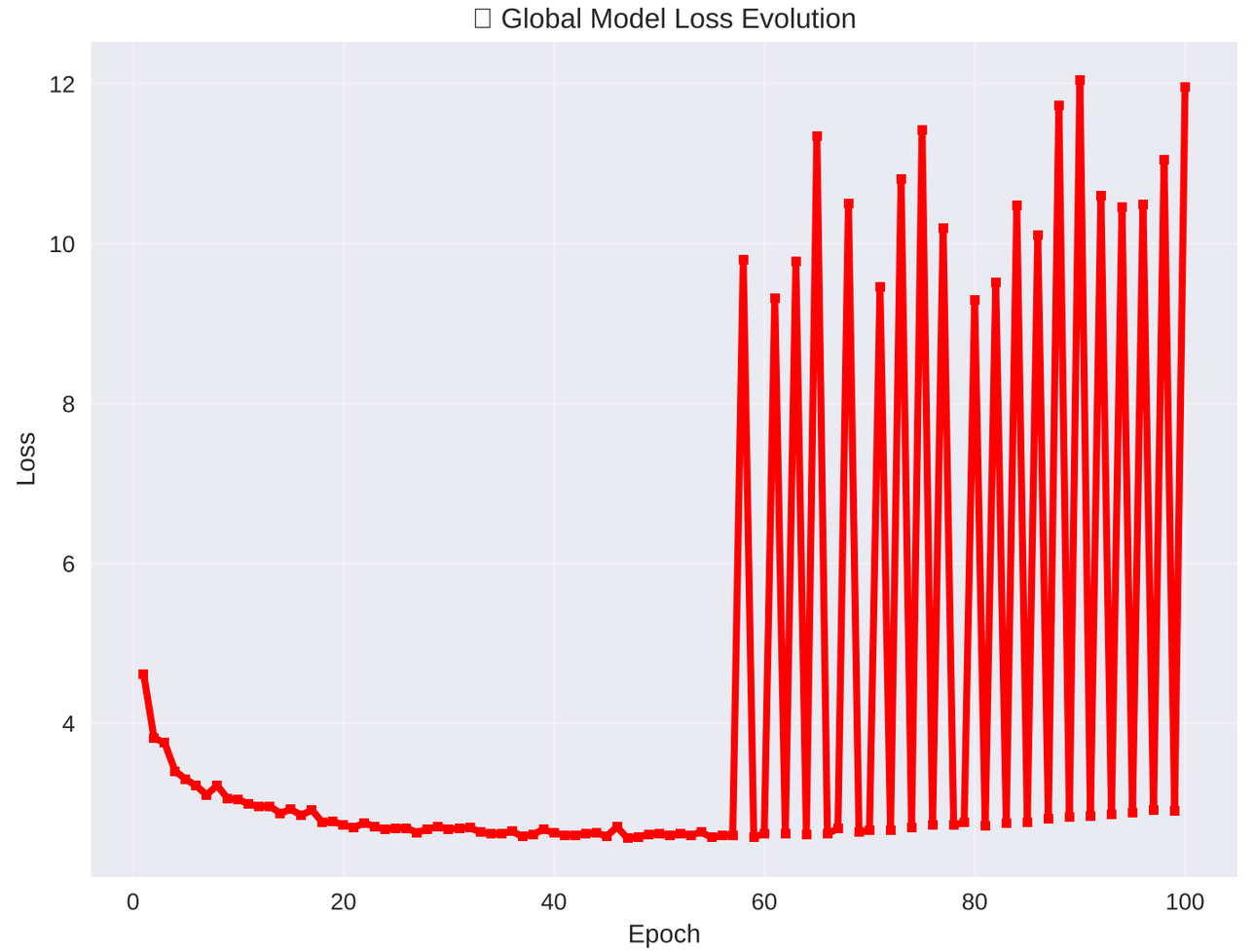
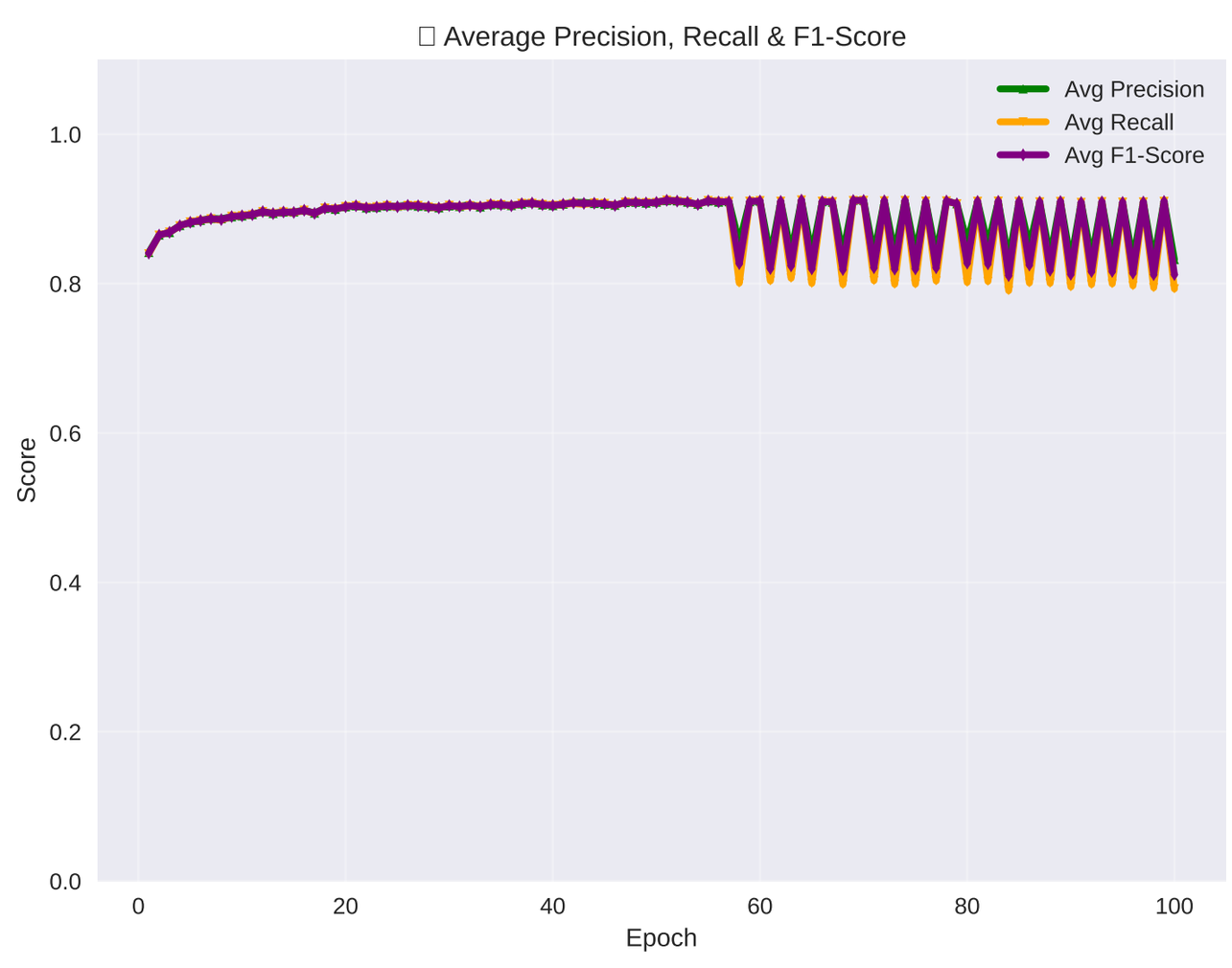
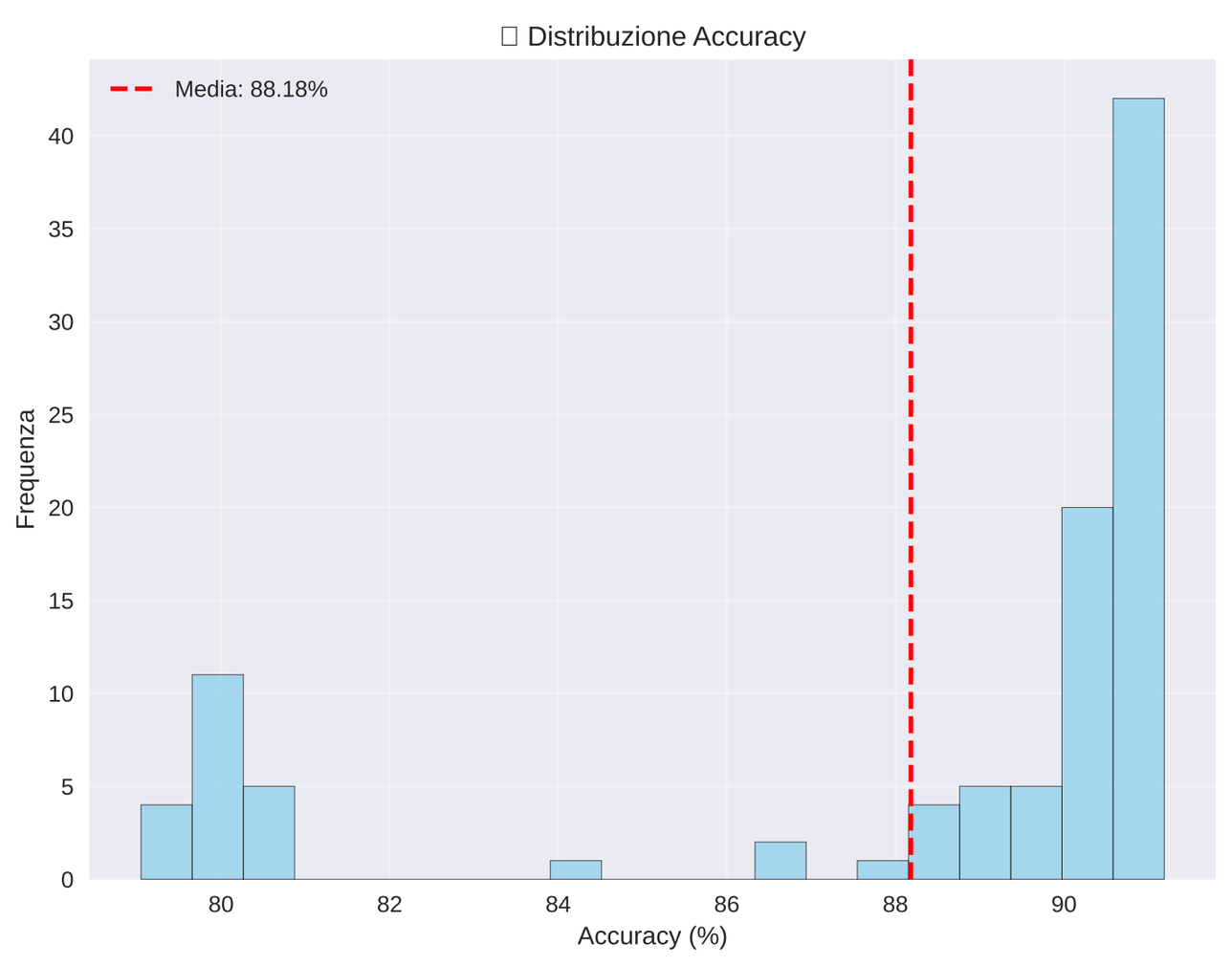
Federated Learning - Main Metrics Overview



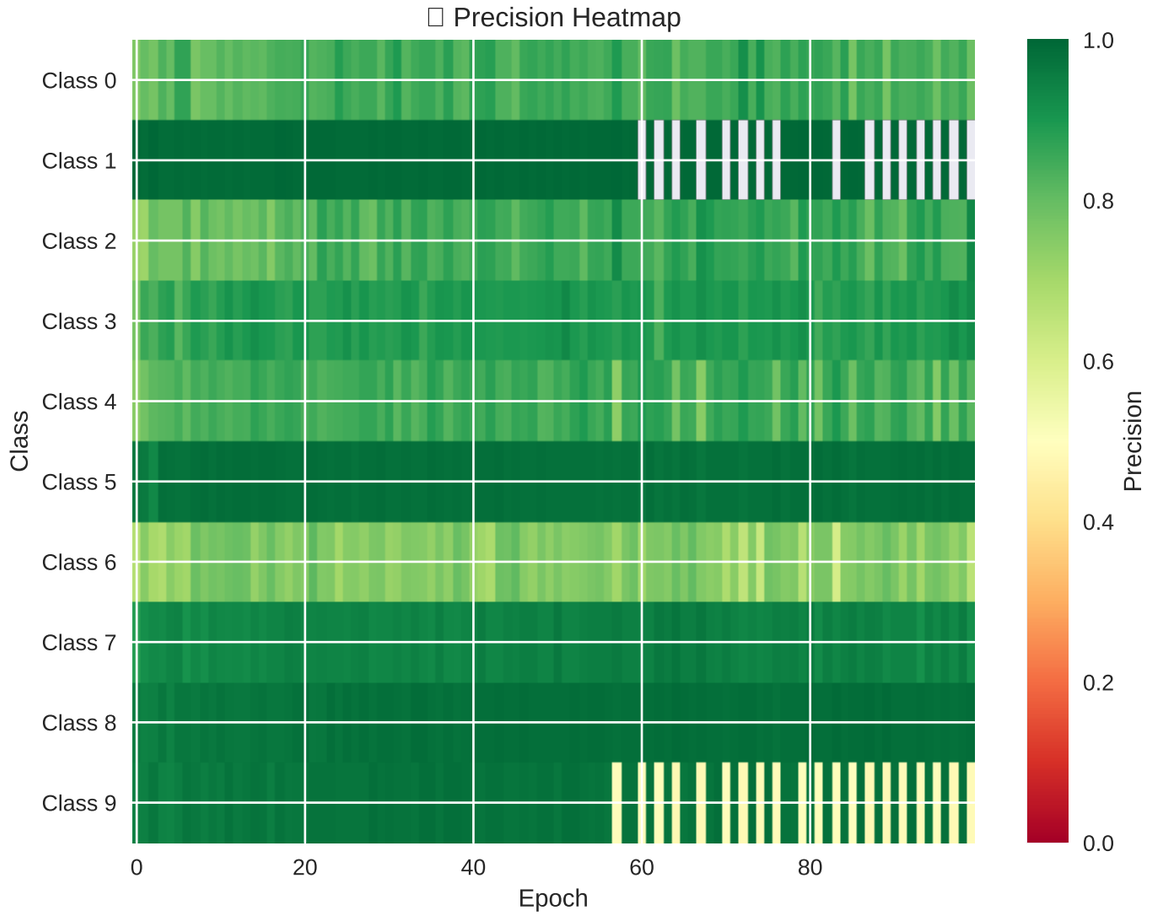


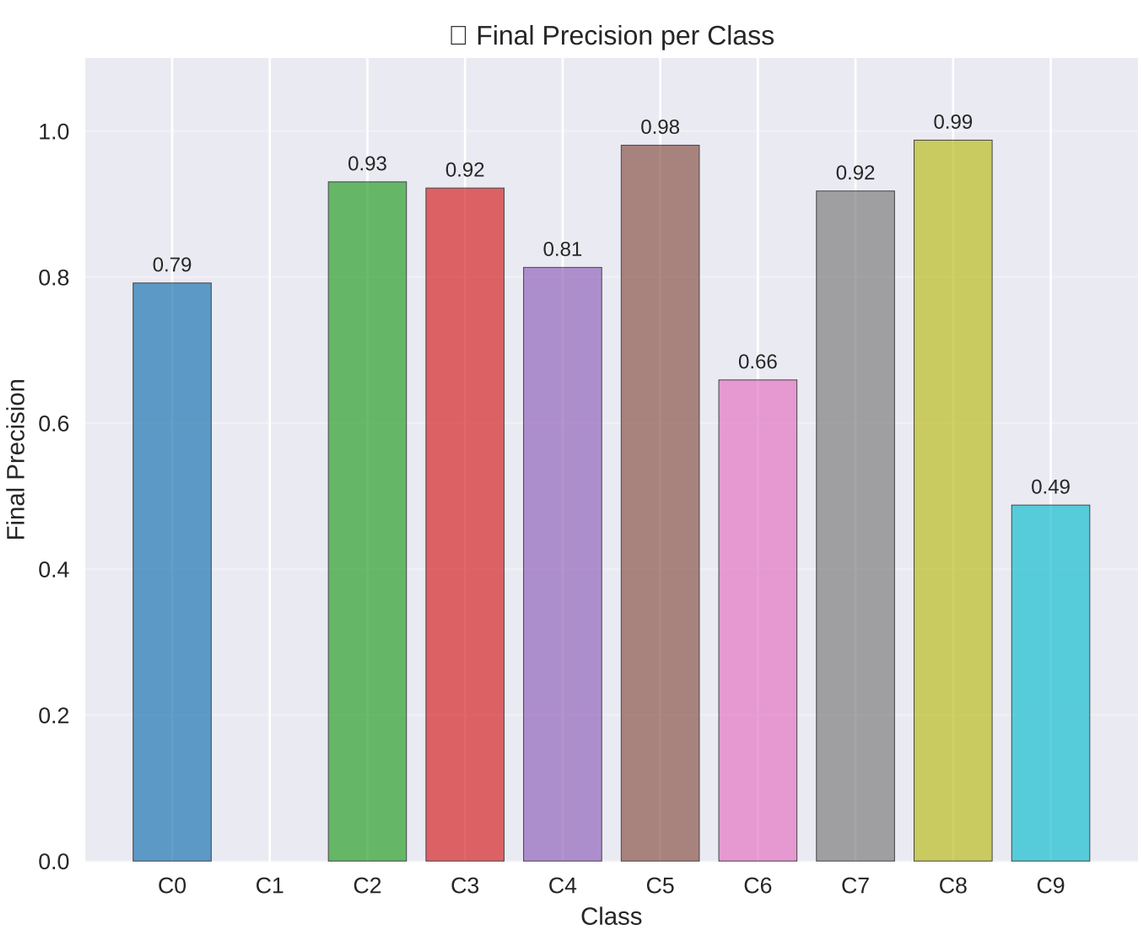


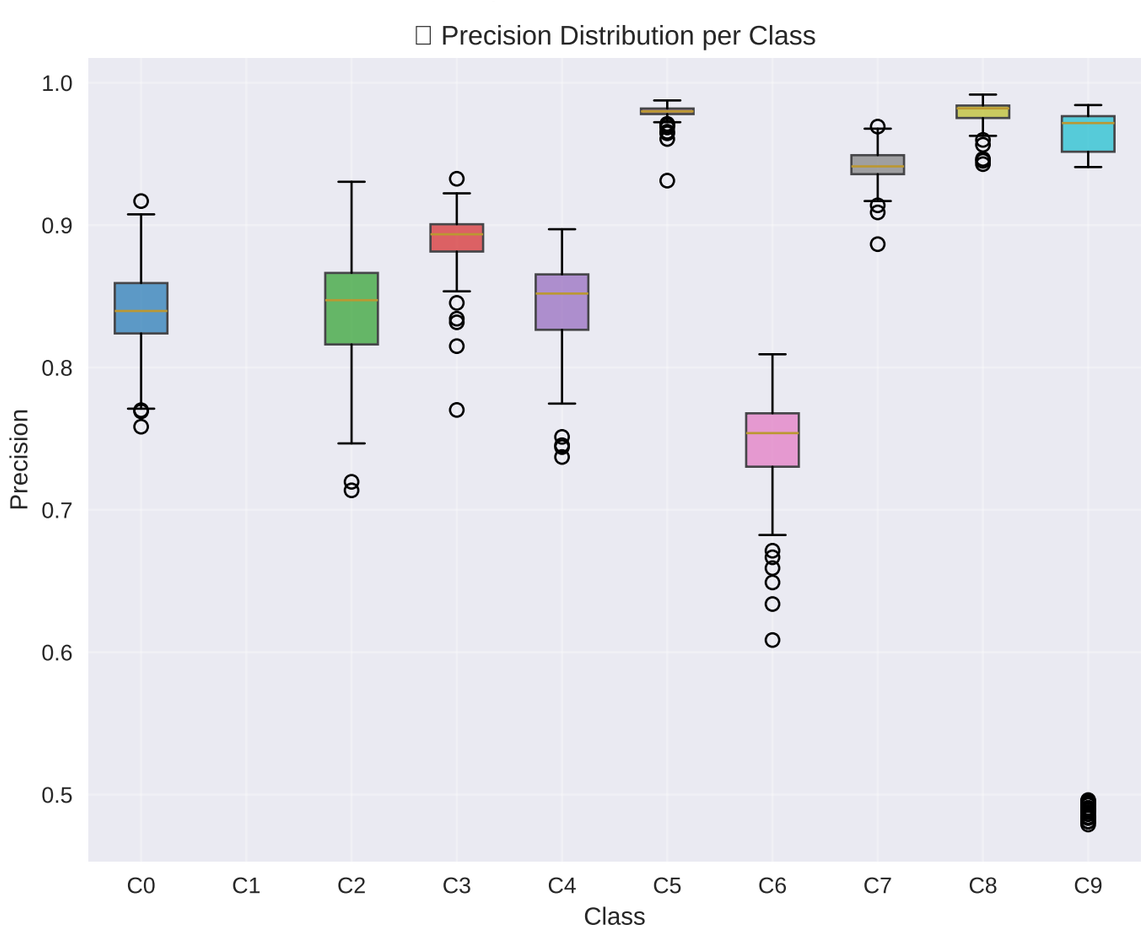


Per-Class Precision Analysis

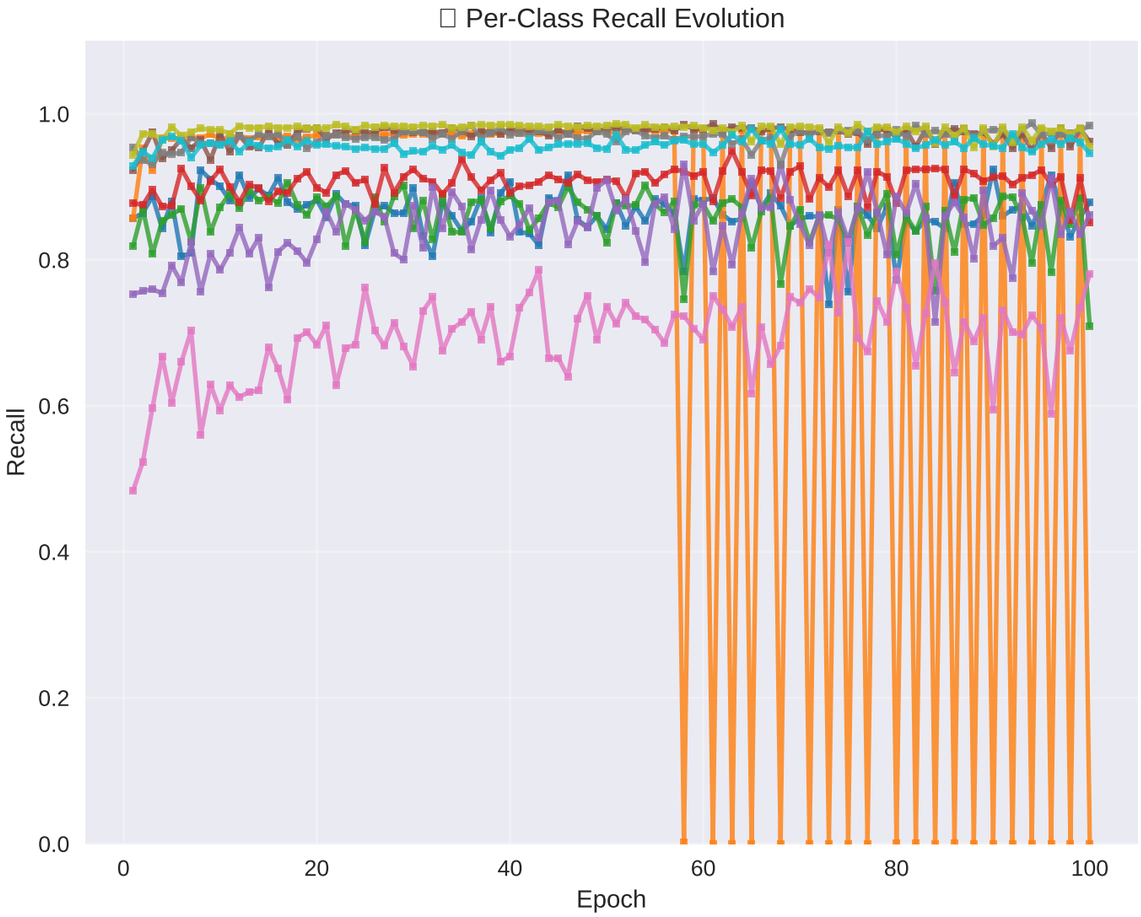


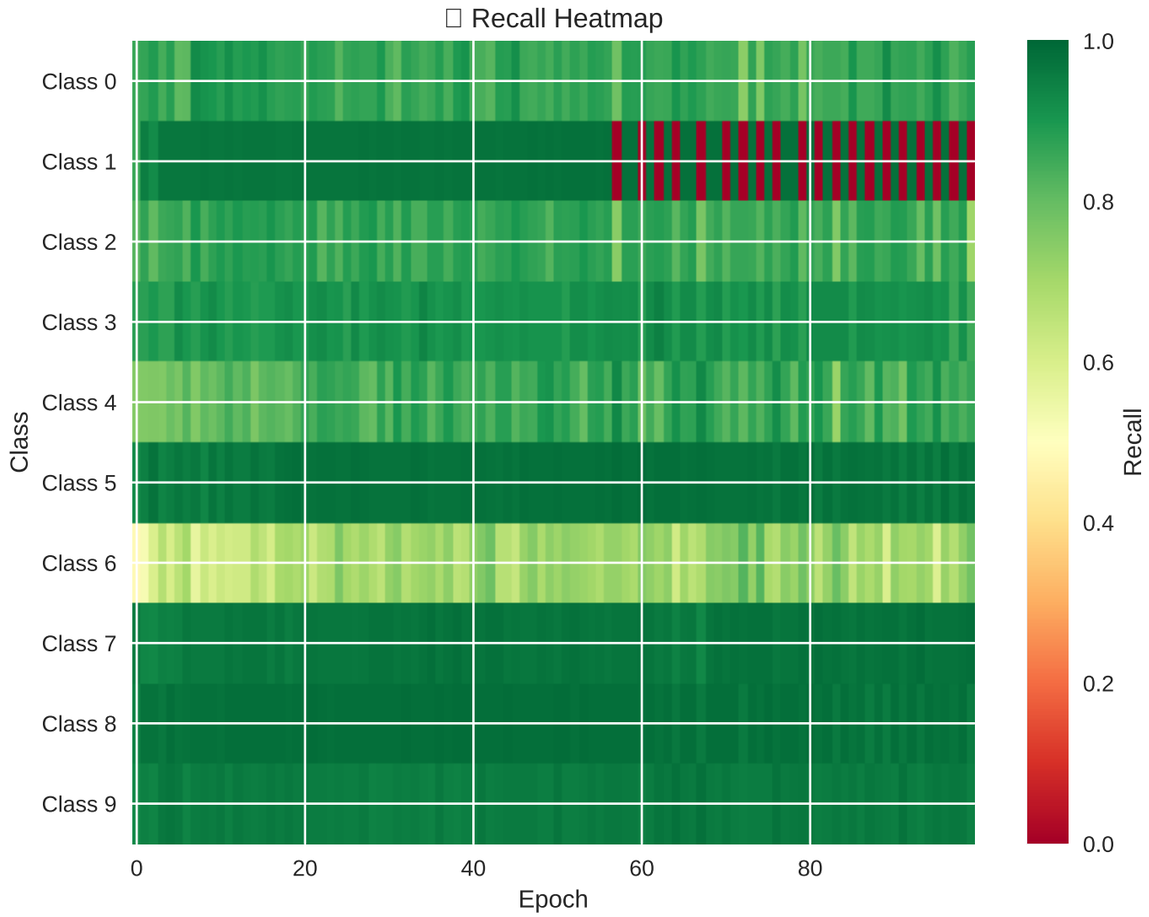


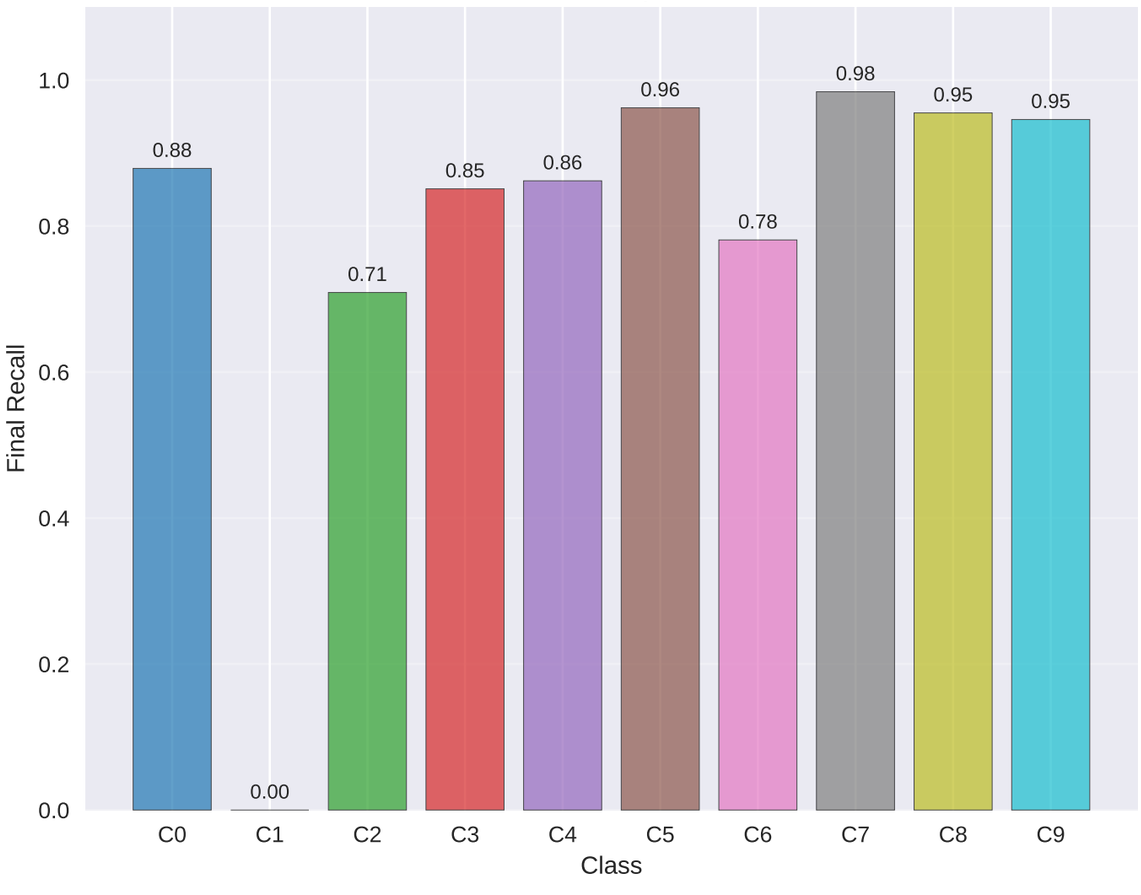


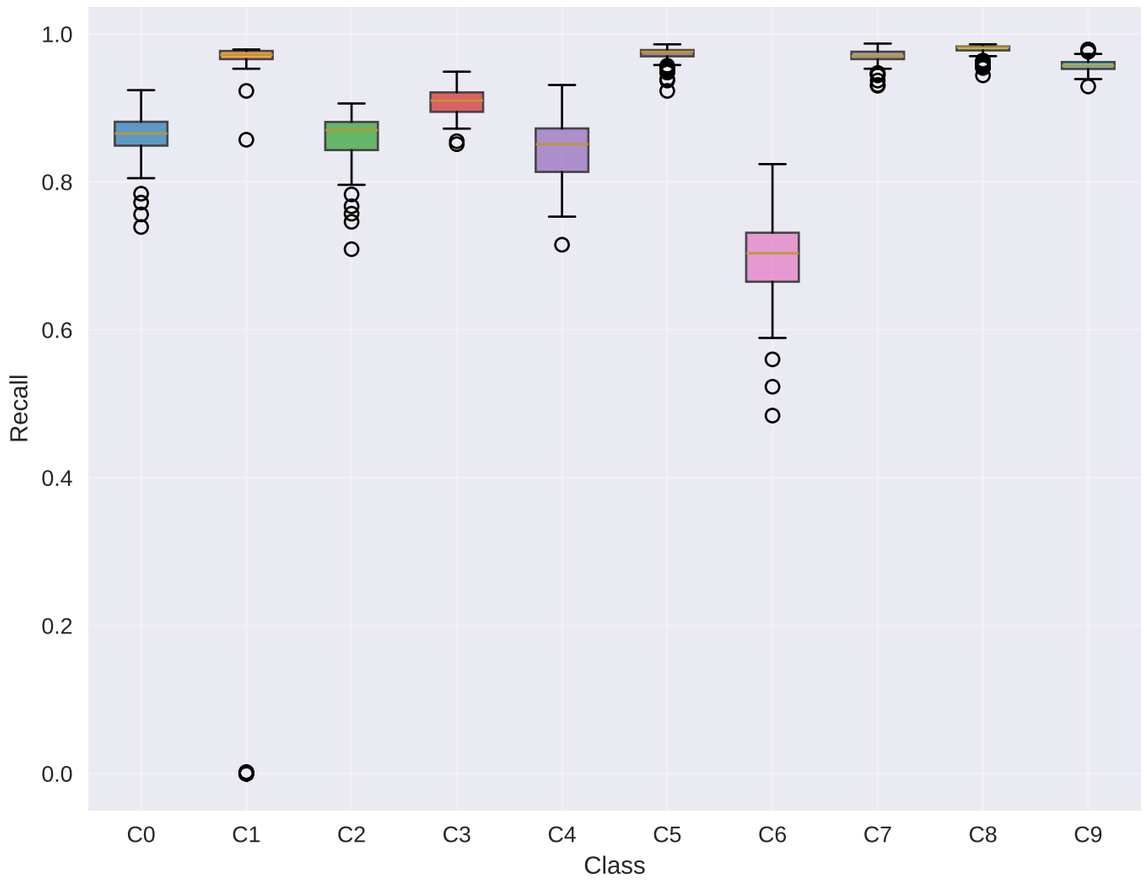


Per-Class Recall Analysis

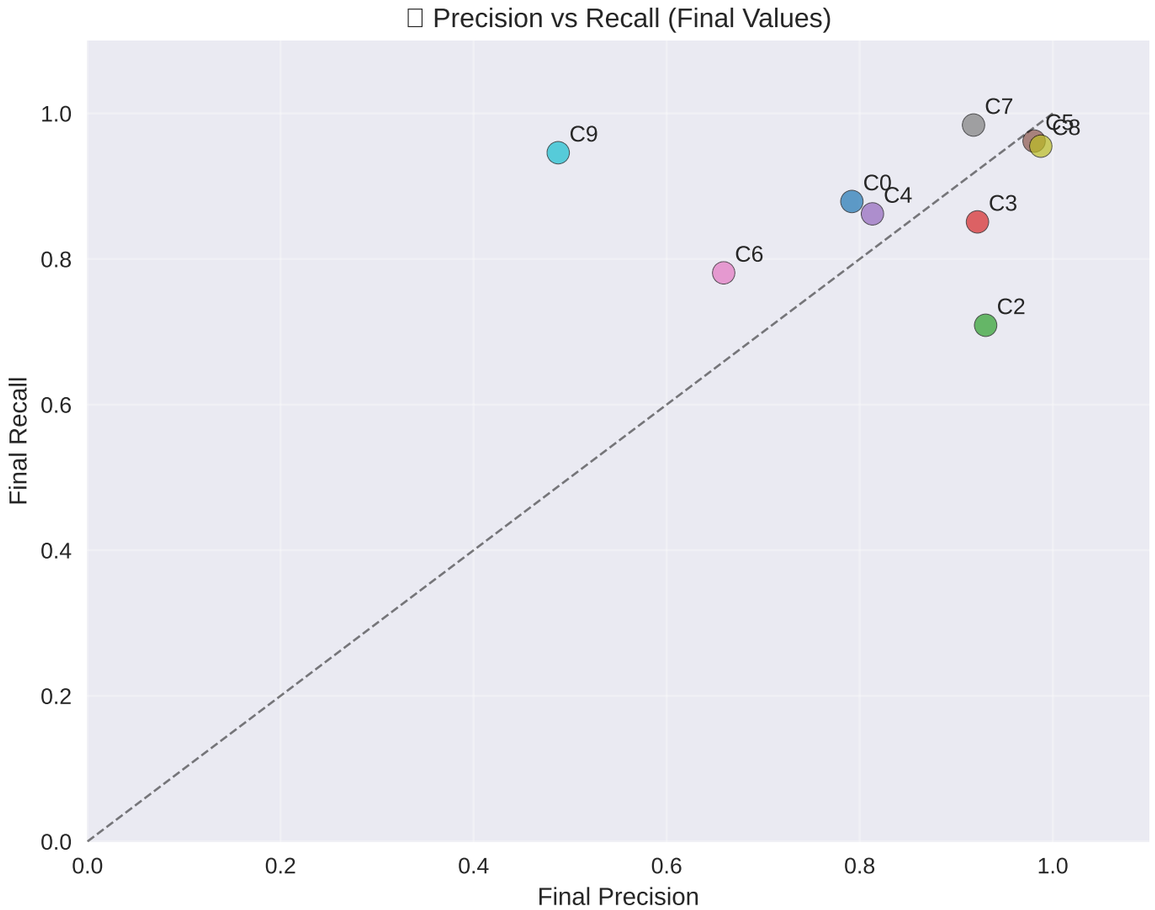


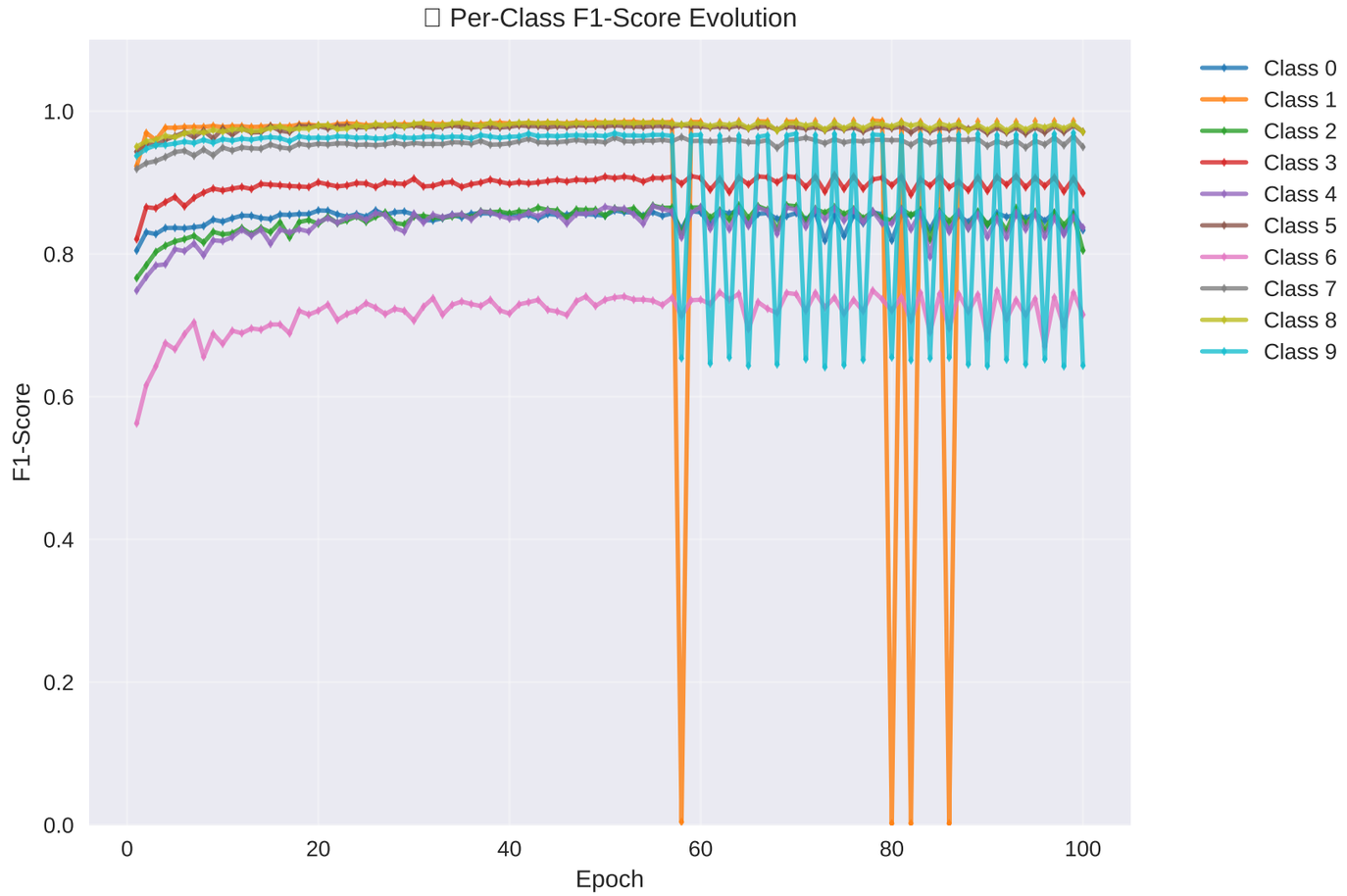




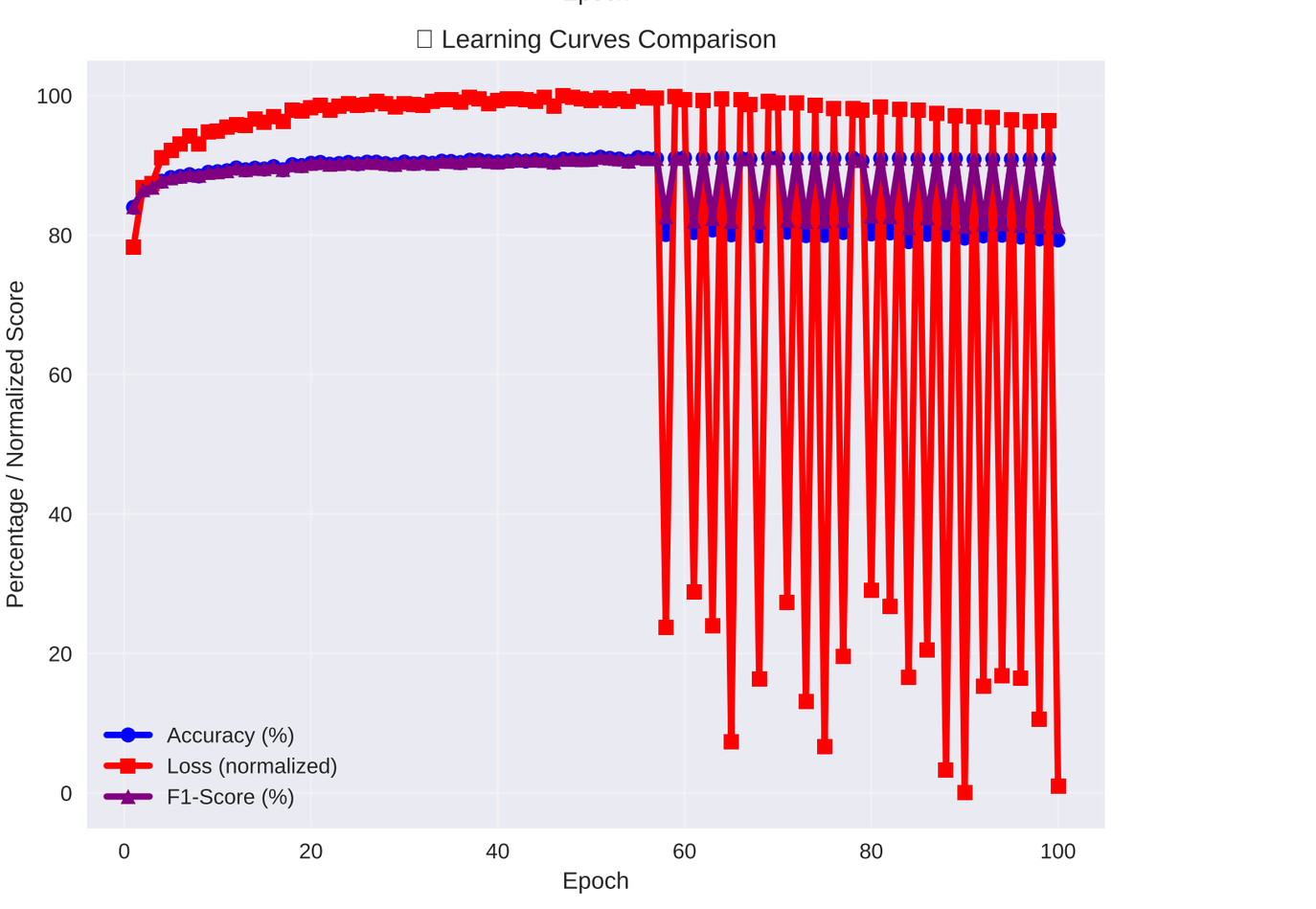


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-04 21:28:37
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 79.29%
• Loss: 11.9594
• Average Precision: 0.832
• Average Recall: 0.793
• Average F1-Score: 0.812

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: -4.69%
• Riduzione Loss: -7.3417
• Stabilità Accuracy (CV): 0.0484

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (nan)
• Peggiore Precision: Classe 1 (nan)
• Migliore Recall: Classe 7 (0.984)
• Peggiore Recall: Classe 1 (0.000)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.792, R=0.879, F1=0.833
• Classe 1: P=nan, R=0.000, F1=nan
• Classe 2: P=0.930, R=0.709, F1=0.805
• Classe 3: P=0.922, R=0.851, F1=0.885
• Classe 4: P=0.813, R=0.862, F1=0.837
• Classe 5: P=0.981, R=0.962, F1=0.971
• Classe 6: P=0.659, R=0.781, F1=0.715
• Classe 7: P=0.918, R=0.984, F1=0.950
• Classe 8: P=0.988, R=0.955, F1=0.971
• Classe 9: P=0.488, R=0.946, F1=0.644

VALUTAZIONE QUALITATIVA:
• Performance Globale: DA MIGLIORARE
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
```